PREPRINT

# Calibrated Critics: Learning When to Trust Rubric Rewards in Open-Ended RL

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**Code** **Model** **Data** **Project**

**Date** October 15, 2026 **Contact** author@example.com

**Abstract**

Rubric-based reinforcement learning has become a standard way to post-train language models on tasks without verifiable answers, yet the judge that scores each rubric item is itself an imperfect proxy. We show that the judge's confidence is informative: rubric items on which the judge is uncertain account for most of the reward that a policy later exploits. We introduce *Calibrated Critics*, which reweights each criterion by an estimate of the judge's reliability on that criterion and prompt. Across long-form research and medical benchmarks, our method improves held-out gold judge scores by up to 6.1 points while matching the baseline's in-domain reward, and it substantially narrows the gap between training and gold judges throughout training.

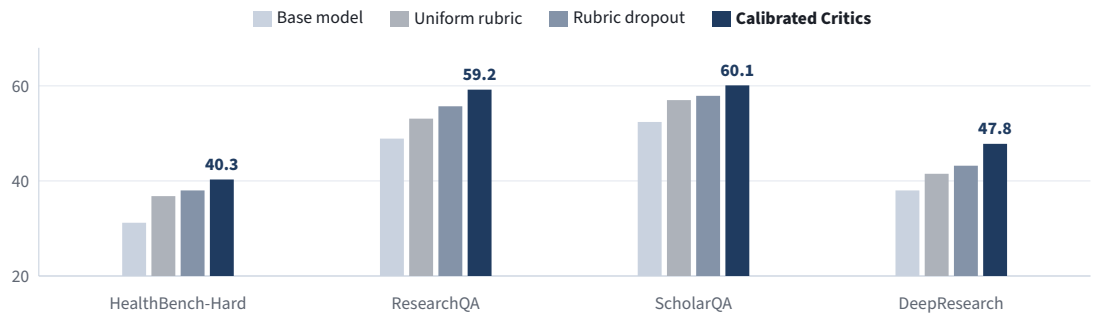



**Figure 1** **Out-of-distribution results.** Gold judge scores on four long-form benchmarks. Calibrated Critics improves over every baseline on all four.

## 1 Introduction

Reinforcement learning with verifiable rewards works well when a ground truth exists (Shao et al., 2024). Many of the tasks we care most about, such as explaining a research area or giving medical advice, are open-ended. A common recipe is to write a rubric for each prompt and let an LLM judge grade each criterion (Gunjal et al., 2025). The rubric, however, is a proxy, and a policy optimized against it long enough learns to exploit the difference (Gao et al., 2023).

Our contributions are as follows:

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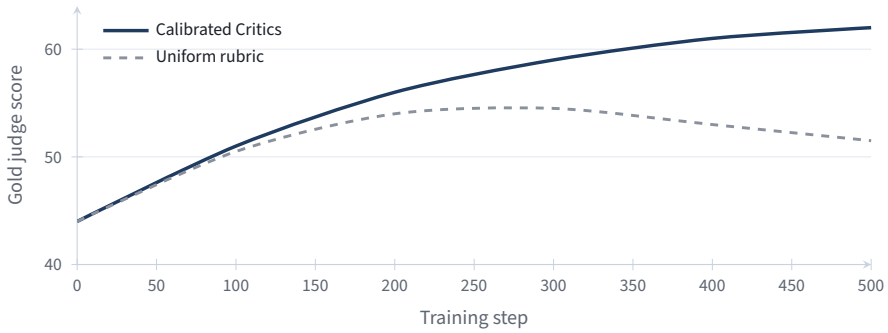



**Figure 2** **Training dynamics.** Gold judge score on held-out prompts during training. The uniform rubric baseline peaks and then declines as the policy exploits unreliable criteria; reliability weighting keeps improving.

- We show that judge uncertainty predicts which rubric items will be exploited, before exploitation happens.
- We propose a per-criterion reliability weight that is cheap to estimate and drops into any GRPO-style objective.
- We evaluate on four long-form benchmarks and two model scales.

**Takeaway 1**

**Judge uncertainty identifies the rubric criteria most vulnerable to reward hacking.** Most exploitation concentrates on a small set of criteria where the judge was already uncertain before policy optimization.

## 2 Method

### 2.1 Reliability-weighted rubric reward

Given a prompt $x$ with rubric $\{c_k\}_{k=1}^{K}$ and weights $w_k$, the standard reward for a response $y$ is the weighted fraction of satisfied criteria. We instead use

$$r(x, y) \;=\; \frac{\sum_{k=1}^{K} w_k \, \rho_k(x) \, \mathbb{I}[\, y \models c_k \,]}{\sum_{k=1}^{K} w_k \, \rho_k(x)}, \tag{1}$$

where $\rho_k(x) \in [0, 1]$ is the estimated reliability of the judge on criterion $c_k$.

**Estimating reliability.** We sample the judge $m$ times per criterion and set $\rho_k$ to one minus the normalized entropy of its votes.

**Discussion 1**

**Relationship to rubric dropout.** Dropout makes a fixed rubric harder to exploit by varying which criteria receive reward. Reliability weighting addresses a complementary failure mode: it reduces the influence of criteria whose judgments are unstable even before the policy begins to optimize against them.

**Table 1** **Main results.** Gold judge scores on out-of-distribution benchmarks. Best values are shown in bold.

| Method | HealthBench-Hard | ResearchQA | ScholarQA | Avg. |
|---|---|---|---|---|
| Base model | 31.2 | 48.9 | 52.4 | 44.2 |
| Uniform rubric | 36.8 | 53.1 | 57.0 | 49.0 |
| Rubric dropout | 38.0 | 55.7 | 57.9 | 50.5 |
| **Calibrated Critics** | **40.3** | **59.2** | **60.1** | **53.2** |

## 3 Experiments

Table 1 reports the main results and Figure 2 shows the training dynamics.

**Practical Guidance 1**

**Calibrate the judge before starting reinforcement learning.** Sample each rubric criterion several times on representative prompts, estimate its agreement, and downweight unstable criteria. Recompute these estimates periodically as the policy distribution shifts.

## References

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, 2023.

Anisha Gunjal, Anthony Wang, Elaine Lau, et al. Rubrics as rewards: Reinforcement learning beyond verifiable domains. *arXiv preprint arXiv:2507.17746*, 2025.

Zhihong Shao, Peiyi Wang, Qihao Zhu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

# Appendix

**A Implementation details** . . . . . **4**
A.1 Training hyperparameters . . . . . 4
A.2 Judge configuration . . . . . 4
**B Prompts** . . . . . **4**
**C Additional results** . . . . . **4**

## A Implementation details

### A.1 Training hyperparameters

We train with GRPO using 8 rollouts per prompt, a learning rate of $10^{-6}$, and a KL coefficient of 0.001 for 500 steps.

### A.2 Judge configuration

The training judge is sampled $m = 5$ times per criterion at temperature 0.7.

## B Prompts

**Rubric judge prompt**

```
You are grading a response against a single rubric criterion.
Question: {question}
Response: {response}
Criterion: {criterion}
Answer YES if the response satisfies the criterion, otherwise NO.
```

## C Additional results

Per-benchmark training curves are shown in the supplementary material.